Point2RBox-v3: self-bootstrapping from point annotations via integrated pseudo-label refinement and utilization

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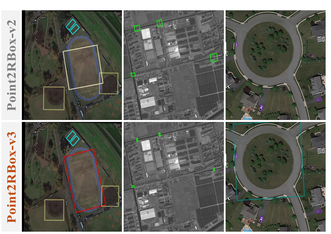
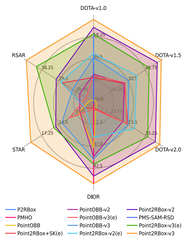
Summary

- (Left) Visual comparisons with the state-of-the-art method Point2RBox-v2. The boxes detected by our method wrap the objects more tightly, with fewer missed detections.
- (Right) Radar plot comparing the performance of our method with 10 other state-of-the-art methods across 6 benchmark datasets. Methods with '(e)' in the legend indicate end-to-end version, while those without '(e)' represent two-stage version.





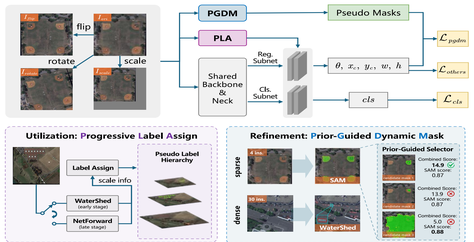
Introduction

- Traditional rotated bounding boxes (RBoxes) provide accurate supervision but are costly: annotating each RBox is 36.5% more expensive than a horizontal box (HBox) and 104.8% more than a point, highlighting the potential of weakly-supervised OOD.
- HBox-based approaches such as H2RBox and H2RBox-v2 have shown strong results, while point-based methods are rapidly advancing.

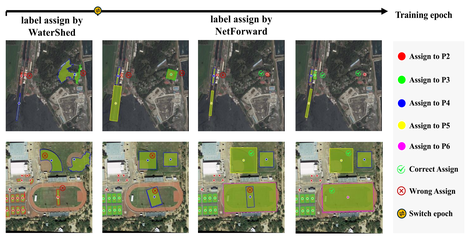
Motivation

- Most existing methods simply distribute labels to the same fpn layer. This wastes the scale information contained in the pseudo labels generated by the model.
- The watershed algorithm exhibits significant performance limitations in sparse scenarios. Can we leverage higher-quality segmentation masks as supervisory signals?

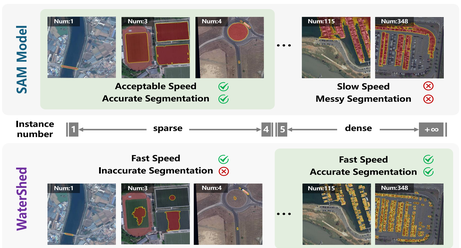
Method



- PLA enables the use of label assignment methods.



- PGDM-Loss overcomes the shortcomings of Watershed in its poor performance in sparse scenes and SAM's poor perfor mance in dense scenes.



Result

Methods	*	PL'	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	AP_{50}
▼ *RBox-supervised OOD*																	
RepPoints	✓	86.7	81.1	41.6	62.0	76.2	56.3	75.7	90.7	80.8	85.3	63.3	66.6	59.1	67.6	33.7	68.45
RetinaNet	✓	88.2	77.0	45.0	69.4	71.5	59.0	74.5	90.8	84.9	79.3	57.3	64.7	62.7	66.5	39.6	68.69
GWD	✓	89.3	75.4	47.8	61.9	79.5	73.8	86.1	90.9	84.5	79.4	55.9	59.7	63.2	71.0	45.4	71.66
FCOS	✓	89.1	76.9	50.1	63.2	79.8	79.8	87.1	90.4	80.8	84.6	59.7	66.3	65.8	71.3	41.7	72.44
S^2A-Net	✓	89.2	83.0	52.5	74.6	78.8	79.2	87.5	90.9	84.9	84.8	61.9	68.0	70.7	71.4	59.8	**75.81**
▼ *HBox-supervised OOD*																	
Sun et al.	×	51.5	38.7	16.1	36.8	29.8	19.2	23.4	83.9	50.6	80.0	18.9	50.2	25.6	28.7	25.5	38.60
BoxInst-RBox[2]	×	68.4	40.8	33.1	32.3	46.9	55.4	56.6	79.5	66.8	82.1	41.2	52.8	52.8	65.0	30.0	53.59
H2RBox	✓	88.5	73.5	40.8	56.9	77.5	65.4	77.9	90.9	83.2	85.3	55.3	62.9	52.4	63.6	43.3	67.82
EIE-Det	✓	87.7	70.2	41.5	60.5	80.7	76.3	86.3	90.9	82.6	84.7	53.1	64.5	58.1	70.4	43.8	70.10
H2RBox-v2	✓	89.0	74.4	50.0	60.5	79.8	75.3	86.9	90.9	85.1	85.0	59.2	63.2	65.2	70.5	49.7	**72.31**
▼ *Point-supervised OOD*																	
Point2Mask-RBox[2]	×	4.0	23.1	3.8	1.3	15.1	1.0	3.3	19.0	1.0	29.1	0.0	9.5	7.4	21.1	7.1	9.72
P2BNet+H2RBox	×	24.7	35.9	7.1	27.9	3.3	12.1	17.5	17.5	0.8	34.0	6.3	49.6	11.6	27.2	18.8	19.63
P2BNet+H2RBox-v2	×	11.0	44.8	14.9	15.4	36.8	16.7	27.8	12.1	1.8	31.2	3.4	50.6	12.6	36.7	12.5	21.87
P2RBox	×	87.8	65.7	15.0	**60.7**	73.0	71.7	78.9	81.5	44.5	81.2	41.2	39.3	45.5	57.5	41.2	59.04
PointOBB	×	26.1	65.7	9.1	59.4	65.8	34.9	29.8	0.5	2.3	16.7	0.6	49.0	21.8	41.0	36.7	30.08
Point2RBox	✓	62.9	64.3	14.4	35.0	28.2	38.9	33.3	25.2	2.2	44.5	3.4	48.1	25.9	45.0	22.6	34.07
Point2RBox+SK	✓	53.3	63.9	3.7	50.9	40.0	39.2	45.7	76.7	10.5	56.1	5.4	49.5	24.2	51.2	33.8	40.27
Point2RBox+SK	×	66.4	59.5	5.2	52.6	54.1	53.9	57.3	90.8	3.2	57.8	6.1	47.4	22.9	55.7	40.5	44.90
PointOBB-v2	×	64.5	27.8	1.9	36.2	58.8	47.2	53.4	90.5	62.2	45.3	12.1	41.7	8.1	43.7	32.0	41.68
PMS-SAM-RSD	✓	69.0	39.5	6.7	44.8	64.7	71.9	79.6	79.8	2.7	60.0	12.1	32.6	39.6	44.8	42.5	46.00
PointOBB-v3	✓	30.9	39.4	13.5	22.7	61.2	7.0	43.1	62.4	59.8	47.3	2.7	45.1	16.8	55.2	11.4	41.29
PointOBB-v3	×	52.9	54.4	21.3	52.7	65.6	44.9	67.8	87.2	26.7	73.4	32.6	53.3	39.0	56.4	10.2	49.24
Point2RBox-v2	✓	78.4	52.7	8.3	40.9	71.0	60.5	74.7	88.7	65.5	72.1	24.4	26.1	30.1	50.7	21.0	51.00
Point2RBox-v2	×	88.0	**72.6**	8.0	46.2	**79.6**	76.3	86.9	89.1	**79.7**	82.9	26.2	45.3	45.8	**66.3**	46.3	62.61
Point2RBox-v2+PLA (ours)	✓	82.3	47.8	23.2	36.0	77.9	75.5	86.6	85.3	72.0	76.5	28.0	33.4	39.4	52.7	31.7	56.55
Point2RBox-v2+PLA (ours)	×	88.3	71.5	26.3	45.9	78.8	**76.9**	87.5	86.9	74.6	83.1	47.7	45.2	49.0	60.7	46.7	64.61
Point2RBox-v2+PLA+CS[1] (ours)	✓	84.6	51.5	25.4	37.2	78.7	75.3	86.5	86.4	75.8	74.9	31.3	33.6	40.2	54.5	34.2	58.00
Point2RBox-v3 (ours)	✓	86.5	53.4	35.7	37.5	78.8	75.3	86.3	86.6	66.0	80.2	29.7	49.9	36.7	59.1	32.5	59.61
Point2RBox-v3 (ours)	×	**89.0**	72.5	**41.6**	45.1	70.4	76.4	**87.7**	85.1	75.7	**84.8**	44.4	**55.4**	45.7	59.8	**48.8**	**66.09**

Methods	*	DOTA-v1.0	DOTA-v1.5	DOTA-v2.0	DIOR	STAR	RSAR
▼ *RBox-supervised OOD*							
RetinaNet	✓	68.69	60.57	47.00	54.96	21.80	57.67
GWD	✓	71.66	63.27	48.87	57.60	25.30	57.80
FCOS	✓	72.44	64.53	51.77	59.83	**28.10**	**66.66**
S^2A-Net	✓	**75.81**	**66.53**	**52.39**	**61.41**	27.30	66.47
▼ *HBox-supervised OOD*							
Sun et al.	×	38.60	-	-	-	-	-
H2RBox	✓	70.05	61.70	48.68	57.80	17.20	49.92
H2RBox-v2	✓	72.31	64.76	50.33	57.64	**27.30**	**65.16**
AFWS	✓	**72.55**	**65.92**	**51.73**	**59.07**	-	-
▼ *Point-supervised OOD*							
P2RBox	×	59.04	-	-	-	-	-
PMHO	×	-	-	-	46.20	-	-
PointOBB	×	30.08	10.66	5.53	37.31	9.19	13.80
Point2RBox+SK	✓	40.27	30.51	23.43	27.34	7.86	27.81
PointOBB-v2	×	41.68	30.59	20.64	39.56	9.00	18.99
PMS-SAM-RSD	✓	46.00	-	-	-	-	-
PointOBB-v3	✓	41.20	31.25	22.82	37.60	11.31	15.84
PointOBB-v3	×	49.24	33.79	23.52	40.18	12.85	22.60
Point2RBox-v2	✓	51.00	39.45	27.11	34.70	7.80	28.60
Point2RBox-v2	×	62.61	54.06	38.79	44.45	14.20	30.90
Point2RBox-v3 (ours)	✓	59.61	48.34	34.47	41.50	14.60	40.80
Point2RBox-v3 (ours)	×	**66.09**	**56.86**	**41.28**	**46.40**	**19.60**	**45.96**

* Comparison tracks: ✓ = End-to-end training and testing; × = Two-stage training with pseudo labels. P2RBox/PMS-SAM-RSD/PMHO/Point2RBox-v3: Pre-trained SAM model. Point2RBox+SK: One-shot sketches for each class.

Generalizability

- Point2RBox-v3 consistently and significantly outperformed the PWOOD baseline.

Method	DOTA-v1.0				DOTA-v1.5			
	10%	20%	30%	full	10%	20%	30%	full
PWOOD	42.35	45.01	49.12	55.87	35.33	41.54	43.02	46.83
PGDM	42.81	47.70	51.02	57.96	37.29	42.02	44.06	49.08
PLA	45.18	50.33	55.67	62.41	40.36	45.24	48.13	53.23
PLA + PGDM	**50.67**	**52.22**	**58.02**	**64.57**	**43.25**	**47.05**	**50.51**	**56.18**

- Point2RBox-v3 shows superior generalization beyond remote sensing, achieving gains of 4.5% on OCDPCB and 8.6% on DIATOM over Point2RBox-v2.

Methods	OCDPCB	DIATOM
Point2RBox-v2	36.4	69.0
Point2RBox-v3	**40.9**	**77.6**

Conclusion

- In point-supervised tasks, both the quality and utilization efficiency of pseudo labels are critical factors that can yield substantial performance gains.
- The label assignment module distributes objects to FPN layers based on their scale-specific receptive fields. Due to the intrinsic spatial approximation of convolutional operations and empirically defined assignment thresholds, this approach exhibits inherent tolerance to pseudo labels imperfections while maintaining detection accuracy.
- It advances the state of the art by a large amount, achieving 66.09%, 56.86%, 41.28%, 46.40%, 19.60% and 45.96% on the DOTA-v1.0, DOTA-v1.5, DOTA-v2.0, DIOR, STAR and RSAR respectively.

Acknowledgment

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- Project Page: https://johnson-magic.github.io/point2rbox-v3.github.io/
- Code: https://github.com/VisionXLab/Point2RBox-v3